The Prevalence of Defective Color Vision in Old World Monkeys and Apes

Gerald H. Jacobs,* Gary A. Williams

Neuroscience Research Institute and Department of Psychology, University of California, Santa Barbara, CA 93106

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Abstract: This article examines the question of whether the congenital red/green color vision defects that are common in human populations can be detected in other catarrhine primates. Spectral sensitivity functions obtained from more than 100 Old World monkeys and apes reveal no hint of such individual variation. A literature survey on the subject leads to the same conclusion and provides the further inference that, if color defects do exist in these nonhuman primates, they must be rare. Two classes of explanation for the apparent low incidence of color defects in Old World monkeys and apes are discussed. © 2000 John Wiley & Sons, Inc. Col Res Appl, 26, S123–S127, 2001

Key words: defective color vision; nonhuman primates; dichromacy; opsin genes

INTRODUCTION

Inherited red/green color vision defects are frequent, affecting as many as 7–8% of all males in some human populations.[1,2] Much has been learned about the biological basis of these defects. Early it was established that these defects are X-chromosome linked, and more recently it has become clear that they are traceable to variations in the cone photopigment complement, in turn reflecting variations in the structure and combinations of genes specifying cone opsins. Studies of opsin genes also provide a scenario for the evolution of trichromatic color vision among catarrhine (Old World) primates. Comparisons of the genes specifying the middle (M) and long (L) wavelength cone opsins of primates show they have high sequence similarity. Judged in comparison with other cone opsin genes, this observation leads to the conclusion that primate M and L pigments diverged about 35–40 million years ago, probably as a result of gene duplication.[3,4] This is at about the time of the divergence of catarrhines and platyrrhines (New World monkeys), but well before the robust radiation of catarrhine primates began.[5] The timing of these events implies that the photopigment basis for trichromatic color vision might be expected to be common to all Old World monkeys and hominoids.

Studies of cone opsin genes have also revealed something of the basic mechanisms underlying red/green color vision deficiencies.[6-8] In addition to their high sequence similarity, the M and L cone-opsin genes lie in a tandem array on the X-chromosome, and these two structural features promote frequent unequal meiotic recombination, thus altering the gene array and the subsequent identity and nature of the M and L pigments. It has been noted that, although the underlying mechanisms and nature of color vision seem very consistent across Old World monkeys and hominoids, there are no obvious indications that the color-vision deficiencies that are such prominent features of vision in our species appear in any other catarrhines.[9-11] If real, this difference has implications for understanding the evolution of primate color vision, so we used a twofold strategy to study it more closely — first, by examining a collection of spectral sensitivity functions equivalently measured on human subjects and a number of other Old World primates and, second, by conducting a literature survey that allows a more systematic evaluation of the prospects for color defective vision in Old World monkeys and apes.

COMPARISON OF SPECTRAL SENSITIVITIES OF OLD WORLD PRIMATES

Photopic spectral sensitivity functions have frequently been used to provide insight into the nature of the underlying cone pigments and, by extension, the prospects for color

* Correspondence to: Gerald H. Jacobs, Neuroscience Research Institute and Department of Psychology, University of California, Santa Barbara, CA 93106 (e-mail: jacobs@psych.uscb.edu)

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vision. Over the past 15 years, we have obtained spectral sensitivity functions from a large sample of Old World monkeys and apes. Here we ask what inferences can be obtained from these results.

Measurement Technique and Sample

Spectral sensitivity functions were measured using a noninvasive electrophysiological technique termed electroretinogram (ERG) flicker photometry. This procedure has been fully described elsewhere.[12,13] A photometric paradigm was used to equate the stimulus-locked ERG responses generated by a 31.25 Hz monochromatic test light and by a 31.25 Hz achromatic reference light. The test and reference lights were interleaved in a pulse train from which 50 responses were averaged for both test and reference stimuli. Over a test session, photometric equations were completed for a wide array of test and reference light combinations and from these results spectral sensitivity functions were derived.

To provide context, results from Old World monkeys and apes are compared to ERG measurements made on human subjects (all males). The color vision of the human subjects was first established using results from standard tests with screening plates (Ishihara and Farnsworth–Munsell 100 Hue) and from subsequent measurements of Rayleigh matches obtained as described elsewhere.[14] ERGs were recorded from human subjects with the procedures outlined above. The only essential differences between the experiments run on human and nonhuman subjects were (a) fiber electrodes were employed for human subjects while contact-lens electrodes were used with nonhuman subjects, and (b) ERGs were recorded from sedated nonhuman subjects; for human subjects, the position of the eye was held fixed through the use of a dental impression bite. Recordings were obtained from a total of 107 Old World monkeys and apes. Included in the sample were common chimpanzees and individuals from eight species of cercopithecine monkey and three species of colobine monkey. Several aspects of the results from these nonhuman primates have been reported.[13,15-17]

Assay for the Presence of Dichromatic Color Vision

We screened for dichromacy using a chromatic adaptation test conducted in the ERG flicker photometric paradigm. Photometric equations were made between 540–630 nm lights for two conditions of chromatic adaptation (540 and 630 nm). A systematic difference in the equations for the latter two conditions indicates a failure of univariance and implies the presence of two spectrally discrete M/L photopigments. Plotted in Fig. 1 is a summary of the shifts in 540/630 nm equations obtained for three groups of subjects. The top panel contains results from 35 human dichromats (16 protanopes, 19 deuteranopes). These subjects show no consistent equation change under the two conditions of adaptation. By contrast, each of the 23 human trichromats and each of a total of 104 Old World monkeys and apes show clear failure of univariance. Note in particular that

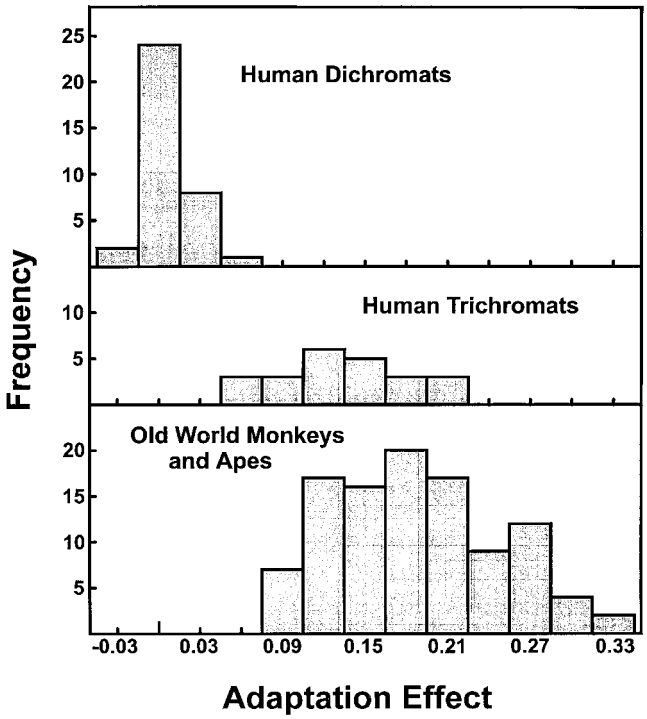


FIG. 1. Summary results from a test designed to establish whether an individual had the photopigment basis for dichromatic or trichromatic color vision. As described in the text, the adaptation effect is the difference in an ERG photometric equation between 540 and 630 nm lights obtained under conditions of concurrent adaptation with red and green lights. The absence of an adaptation effect (value ≈ 0.0) indicates the presence of only a single M/L photopigment and dichromatic color vision. Conversely, a positive value predicts the presence of trichromatic color vision. Bin width = 0.03 log unit.

there is no overlap in the results obtained from the nonhuman primates and known dichromats. It seems clear that all the nonhuman primates in this sample have the requisite photopigment basis for trichromatic color vision.

Spectral Sensitivity Functions

Complete spectral sensitivity functions were obtained from each of the subjects described above. Averages were taken across the two groups. To avoid any potential complications from females who might be heterozygous at their X-chromosome opsin gene sites, the sample of Old World monkeys and apes was restricted to males ($N = 85$). Figure 2 (left) shows the mean spectral sensitivity values obtained from the two samples (± 1 SD). Prior to averaging, all the individual functions were scaled, so that they had the same average height on the sensitivity axis. As can be seen, the functions for the two groups are quite similar in shape. On average, the nonhuman primates are slightly more sensitive to short wavelengths and less sensitive to long wavelengths. Levene's test for homogeneity of variance revealed that only one of 22 test wavelengths differed significantly for the two groups, $F(1,111) = 4.736$, $p < 0.05$. Individual vari-

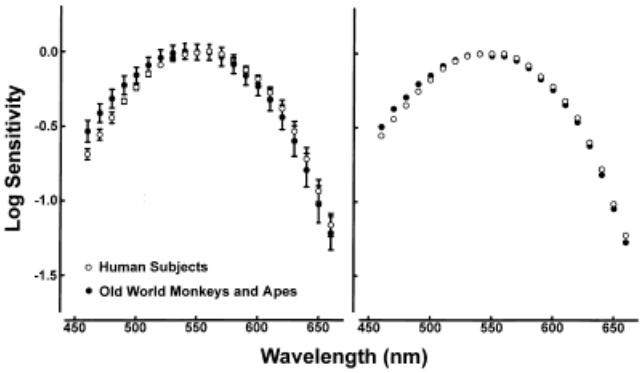


FIG. 2. Averaged spectral sensitivity functions obtained from 23 male normal human trichromats and 85 male Old World monkeys and apes. Each function was measured using ERG flicker photometry. Prior to the averaging plotted in the left panel, the function for each individual was adjusted so that all the individual functions had the same average height on the sensitivity axis. Error bars: ± 1 SD. In the figure on the right, the averaged functions for the two groups have been corrected for lens absorption.

ability in sensitivity within each group is not obviously different.

Large variations in the spectral positioning of the M and L cone pigments produce significant changes in the shapes of ERG-based spectral sensitivity functions, e.g., human anomalous trichromats have ERG spectral sensitivity functions that markedly differ from those of normal trichromats. There are no indications of variation in the nonhuman sample of that magnitude. In addition to variations in M/L pigment positioning, there is a well-known list of other factors that can influence the shapes of spectral sensitivity functions; for example, the proportions of the two pigment types in the retina and significant variations in density spectra of ocular filters. It appears likely that human lens density differs from that of the other catarrhines. Thus, in the right panel of Fig. 2, the averaged functions have been corrected for lens density. Standard corrections were used for the human data; for the nonhuman dataset, it was arbitrarily assumed that preretinal absorption in all these species may be similar to measurements made on macaque eyes.[18] With that correction, the human and nonhuman primate datasets are brought into reasonable agreement. At the very least, these comparisons suggest that the trichromatic color vision of all these species is likely to be very similar.

LITERATURE SURVEY ON NONHUMAN PRIMATE TRICHROMACY

The authors are not aware of any published claims documenting the discovery of an Old World monkey or ape having color-defective vision. As an alternative, we thought it might be possible to set some limits for the potential incidence of color-vision defect by conducting a literature survey to determine the number of animals that had been studied sufficiently to allow the positive diagnosis of trichromacy. To do this, a search was made of the Medline database using the keywords "Monkey Color Vision." This database extends from 1965 to the present and the search yielded in excess of 600 references. This listing was supplemented by other articles known to us that were either published before the 1965 cutoff or were from periodicals not represented in this database. From this set, a total of 87 studies were judged to contain results that seemed adequate to permit a simple categorization of whether or not the animals studied had trichromatic or dichromatic color vision. Table I provides a breakdown of the number of studies and subjects involved for each of the four principal types of techniques used to infer trichromacy. This exercise yielded a total of 573 subjects, all of whom were judged to be trichromats. The sex of the subject was not specified for 31% of the sample. For those cases where it was specified, males outnumbered females by about four to one (almost exactly the male/female ratio in the sample of animals used in the ERG experiment above). Assuming that this same degree of bias holds for those cases where sex was not specified yields an estimated 452 males and 131 females in the total sample. Macaque monkeys were the most frequent targets of examination (75.6%); the remainder was composed of representatives from a variety of cercopithecine species and a number of apes (8.9% of the total sample).

The prevalence of dichromacy among human males averages about 2% in various European populations.[2] Assuming that incidence, and using the Poisson approximation to the binomial theorem, the probability of failing to detect any dichromats in our sample of 452 males is 1.2×10^{-4}. The same approach can be used to estimate the maximum incidence of dichromacy in Old World monkeys and apes given the sample size examined. The value so obtained ($p = 0.05$) is 0.65%. This survey suggests that dichromatic color vision must occur only rarely among Old World monkeys and apes.

DISCUSSION

Examination of carefully measured spectral sensitivity functions and a survey of the literature reveals no evidence that variations from the trichromatic norm that are so obvious in human populations occur among Old World monkey and apes. Although it is, of course, impossible to reject the idea that there are such color-vision defects, the survey implies that the frequency of such variation must be (at least) quite low. What could account for this apparent infrequency?

A classical argument to explain the variations in the incidence of defective color vision in human populations is

TABLE I. Literature survey on the presence of trichromatic color vision in Old World Monkeys and apes.

Principal technique employed	Number of studies	Number of subjects
Gross potential recording	23	178
Behavioral tests	54	311
Outer segment measurements	5	44
Opsin molecular biology	5	40

that these variations index different degrees of relaxation from a formerly strong selection against color blindness.[19,20] The notion is that primitive societies were heavily dependent on hunting and food gathering, circumstances where the color blind might be expected to be at a decided disadvantage and so subject to strong selective pressures. In this view, the changes toward modern civilizations have lessened the dependence on color discriminations, hence weakening selection against color defect. The selectionist argument has been variously criticized, because the more serious color defects (the dichromacies) that should be the most obvious targets for selective pressure may actually not vary in incidence among some population groups where variation would be expected,[21] because there is evidence that defective color vision does in fact provide significant risk factors in contemporary life and thus selection may not be weakened,[22] and because assumptions about the intensity of selection required to account for the differences in the incidence of color blindness seem improbable.[1] Whatever the merit of those particular criticisms, a similar selectionist explanation might be used to explain the effective absence of significant color vision variations among Old World monkeys and apes. Against this possibility we only note that the more than eighty different species that make up this group represent a range of highly variant lifestyles and habitats; perhaps most obviously, some species are principally frugivores, others are principally folivores. At the least it seems reasonable to suppose that trichromatic color discriminations might be much more routinely important for some species than for others and, hence, one might not expect the same stringent selection against color-vision defects in all these species.

Unequal recombination between the M and L opsin genes results in the deletion of genes and in the formation of hybrid or fusion genes, both of which can produce human color vision defects.[6] Another question, then, is whether there might be variation in the structural features of primate M and L opsin genes to account for differences in the incidence of color-vision defect. Three different features have drawn some attention.

(1) People frequently have more than one copy of the M-cone opsin gene[3] and this might enhance the likelihood of unequal cross-overs, fusion gene products, and a consequent increase in the frequency of color defect. There is evidence that is, in fact, the case in human populations: Japanese males are more likely than Caucasian males to have only a single M-opsin gene and they display a lower incidence of defective color vision defect.[23,24] Do Old World monkeys and apes have multiple copies of the M-opsin gene? Apparently so, at least in the case of the several Old World monkeys and chimpanzees that have been so far examined.[25,26] To the extent that variation in gene copy number sets the stage for the production of aberrant gene arrays and consequent color defects, the situation does not at this point appear to be different for humans, Old World monkeys, and apes.

(2) Since the first descriptions of the M and L opsin genes, it has been known that they differ in size due to a shortened intron 1 in the M-opsin gene sequence.[3] Some individuals also have a shortened L sequence intron 1 making the two opsin genes equal in length. The frequency of this variation differs among population groups from less than 1% among Caucasians to about 35% among African Americans.[23] The variation of intron 1 size is inconsistent across the nonhuman primates that have been examined, with long introns in both M and L for gorillas and chimpanzees and short introns in M and L for both orangutans and two species of cercopithecine monkey.[27] Although intron size could influence the likely locations of unequal crossovers, this variation does not seem to obviously correlate with rates of color vision defect.

(3) Much attention has been paid to opsin gene sequence variations in humans and, to a considerably lesser extent, in a number of other nonhuman primates. A number of polymorphic sites have been identified in these genes, some shared between human and nonhuman subjects, others unique to species or groups of species.[26,28] Among human populations, the frequency of one of the polymorphisms at a site crucial for spectral tuning varies in a way that seems correlated with the incidence of color defect. Serine/alanine substitutions at position 180 in the L pigment are associated with 4–7 nm shifts in the spectral peak of the pigment. Among Caucasian males the incidence of serine has been measured at 52–62%.[8,28] The representation of ser180 is considerably higher in Japanese and African American males, estimated at 78–84%.[24,28] Both of the latter groups have a significantly lower incidence of color vision defect than do Caucasians. Although it still entails only a small number of animals, *all* the apes and Old World monkeys that have been examined have been observed to have serine at position 180.[26,28] They thus more closely resemble those human populations having lowered incidence of color defect. Further examination of this apparent correlation seems warranted.

CONCLUSION

The early-stage biological mechanisms underlying color vision seem very similar for all the catarrhine primates. Despite that, the evidence summarized here suggests that color-vision defects that are comparatively common among human populations must have a low incidence among Old World monkeys and apes. The reasons for this difference remain obscure. The M/L opsin gene arrays do not seem to differ among the primates in a fashion that would obviate the possibility of defects in Old World monkeys and apes and, although there is good evidence that selection can serve as a shaping influence on primate photopigment systems,[29] there is reason to be cautious about assuming that such influences are by themselves responsible for effectively eliminating color defects in Old World monkeys and apes.

ADDENDUM: NEW SURVEY OF MACAQUE OPSIN GENES

After this article was accepted for publication, a brief report of a survey of macaque opsin genes was published.[30] It

bears so centrally on the topic of this article that it warrants mention. In a truly Herculean effort, regions in exon 5 known to be critical for distinguishing the M and L versions of the X-linked opsins genes were screened for a total of more than 3100 macaque monkeys. Of the 1199 males in the sample, 3 were found to have only a single X-chromosome opsin gene (an M gene in each case) and should, therefore, have dichromatic color vision. Although the observed frequency of dichromacy in this sample (0.003%) is considerably lower than the upper bound estimated from the literature survey described above, the study provides overwhelming support for the conclusion that color defect is extremely rare among Old World monkeys and apes. There are two other interesting observations made in this report. The three putative color-defective monkeys were all from a small sample of animals (containing 26 males) drawn from a single locale in the Pangandaran National Park of Indonesia. This suggests the possibility that these color-defective macaques are a reflection of a founder effect for an opsin gene defect that has occurred in a restricted population of monkeys. It would be interesting to know more about this population of monkeys and to expand its examination. A second point is that, out of 129 males examined, only 5% were found to have more than one copy of the M opsin gene. This supports the possibility that a difference in gene copy number may be correlated with the low incidence of color defect in trichromatic nonhuman primates and runs counter to the results reviewed above that found multiple gene copies to be common among Old World monkeys and apes.

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